

CODEALPHA INTERNSHIP PROGRAM

Mechanical Design & Simulation

TASK 3: MOTION SIMULATION

Planetary Gear Mechanism

Motion Simulation & Kinematic Analysis Report

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1. Introduction

This report presents the design, assembly, and motion simulation of a Planetary Gear Mechanism created using Creo Parametric. Planetary gear systems are widely used in automotive transmissions, industrial machinery, robotics, and aerospace drives due to their compact structure, high torque density, and multi-stage speed reduction capability.

The objective of this task was to build a fully constrained planetary gear assembly in Creo Parametric and perform a comprehensive motion simulation to analyze the kinematic behavior of the mechanism — including angular position, velocity, and acceleration profiles of the key components.

2. Planetary Gear Mechanism — Overview

2.1 What is a Planetary Gear System?

A planetary gear system consists of one or more outer gears (planet gears) revolving around a central gear (sun gear). The entire assembly is enclosed within a ring gear (annulus). A carrier connects the planet gear axes and rotates as a unit, transmitting output torque.

2.2 Parts Modeled in This Task

Part File	Description
main_gear.prt	Sun Gear — central driving gear (×3 copies used in assembly as planet gears)
ring_gear.prt	Ring Gear (Annulus) — outermost internally toothed fixed gear
runner_gear.prt	Carrier / Runner — structural plate connecting and supporting planet gear axes
gear_connector.prt	Gear Connector — axle/shaft component linking gears within the assembly

Note: The main_gear.prt was instanced three times in the assembly to represent the three equally spaced planet gears meshing with both the sun gear and the ring gear.

3. Component Design & Modeling

3.1 Main Gear (Sun / Planet Gear)

The `maingear.prt` was designed as a standard spur gear with involute tooth profile. This part was reused three times in the assembly as the planet gears, equally distributed at 120° intervals around the central axis. The gear geometry was parameterized to ensure proper meshing with both the ring gear and the sun gear.

3.2 Ring Gear (Annulus)

The `ring_gear.prt` represents the outermost internally toothed gear. In this simulation, the ring gear is held fixed (zero angular velocity), serving as the reaction member. The internal teeth mesh with the three planet gears simultaneously.

3.3 Runner / Carrier

The `runner_gear.prt` acts as the planet carrier. It maintains the positional relationship between the three planet gears, allowing them to revolve around the central axis while simultaneously rotating about their own axes. The carrier is the output member in this configuration.

3.4 Gear Connector

The `gear_connector.prt` serves as the connecting axle/shaft component that links the gear components to the carrier structure and ensures proper transmission of rotational motion through the assembly.

4. Assembly in Creo Parametric

The planetary gear assembly was created by importing all individual part files into a Creo assembly (.asm) file. The following constraints and relationships were defined:

- Ring gear was fully constrained and fixed at the assembly datum (zero degrees of freedom).
- The three planet gear instances (`main_gear.prt` × 3) were placed at 120° angular intervals around the central axis using pattern constraints.
- The `gear_connector.prt` was used to link the planet gears to the carrier (`runner_gear.prt`).
- Gear mesh connections were defined between: Sun gear ↔ Planet gears, and Planet gears ↔ Ring gear.
- A rotational motion driver was applied to the sun gear to initiate the mechanism.

The assembly was verified for interference using Creo's Global Interference Check prior to running the motion simulation.

5. Motion Simulation Setup

5.1 Simulation Tool

Motion simulation was performed using Creo Mechanism (MDO — Mechanism Design Option) integrated within Creo Parametric. The mechanism was defined with appropriate joints, gear pairs, and motion inputs.

5.2 Simulation Parameters

Parameter	Value / Description
Simulation Type	Kinematic Analysis
Motion Input	Constant angular velocity applied to sun gear (input)
Fixed Member	Ring Gear (Annulus) — grounded
Output Member	Planet Carrier (runner_gear)
Duration	10 sec
Time Steps	100
Analysis Mode	Position, Velocity, Acceleration

5.3 Gear Pair Definitions

Gear pair connections were established in the Creo Mechanism module using the Gear Pair tool. The pitch circle radii and gear ratios were entered based on the modeled tooth geometry to ensure accurate kinematic simulation.

6. Simulation Results & Analysis

The motion simulation was successfully executed and the following kinematic graphs were generated using Creo Mechanism's analysis output. The results cover angular position, angular velocity, and angular acceleration of the key components over the simulation duration.

6.1 Position Analysis

The position graph illustrates the angular displacement of the planet carrier (output) and the planet gears over time. The carrier undergoes a steady, uniform angular displacement proportional to the input sun gear rotation, confirming correct gear meshing behavior.

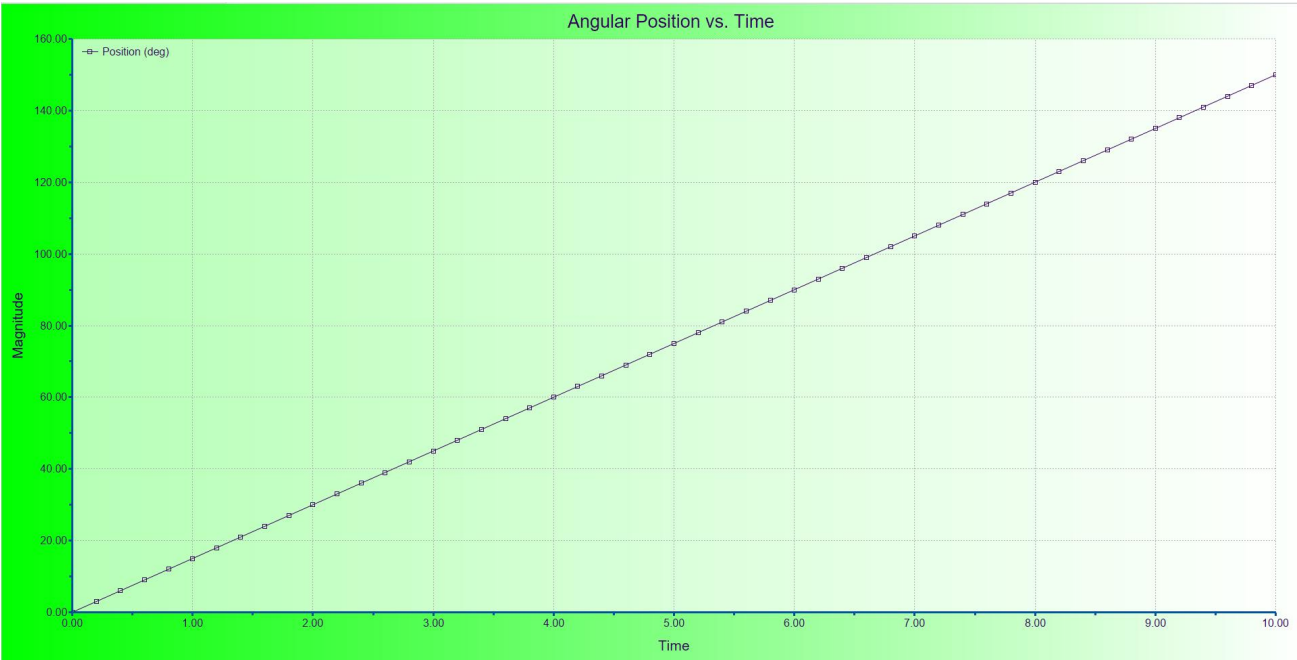


Figure 1: Angular Position vs. Time Graph (Generated from Creo Mechanism)

6.2 Velocity Analysis

The velocity graph shows the angular velocity of the planet carrier relative to the sun gear input speed. The ratio between input and output velocities confirms the gear reduction ratio of the planetary system. The planet gears exhibit compound motion — rotating about their own axes while revolving around the sun gear.

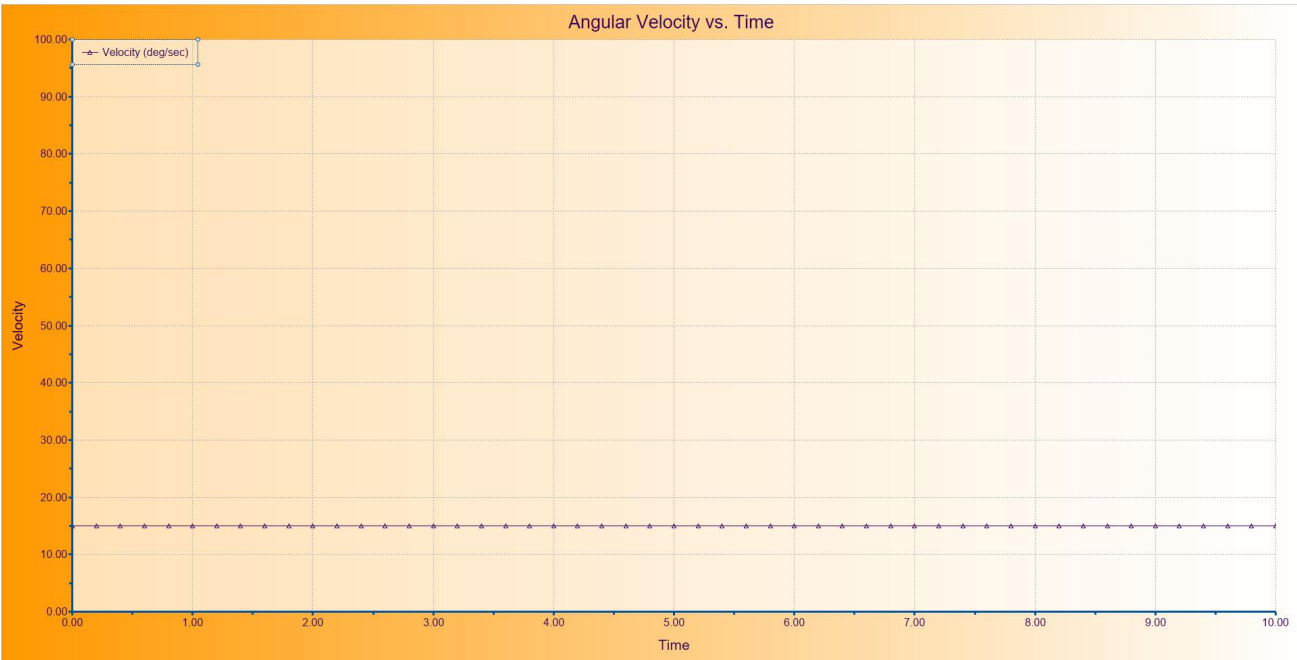


Figure 2: Angular Velocity vs. Time Graph (Generated from Creo Mechanism)

6.3 Acceleration Analysis

The acceleration graph presents the angular acceleration profiles of the mechanism components. Since a constant-velocity motion driver was applied to the sun gear, the system ideally shows near-zero acceleration at steady state. Any transient acceleration at the start of motion is consistent with the ramp-up behavior of the mechanism.

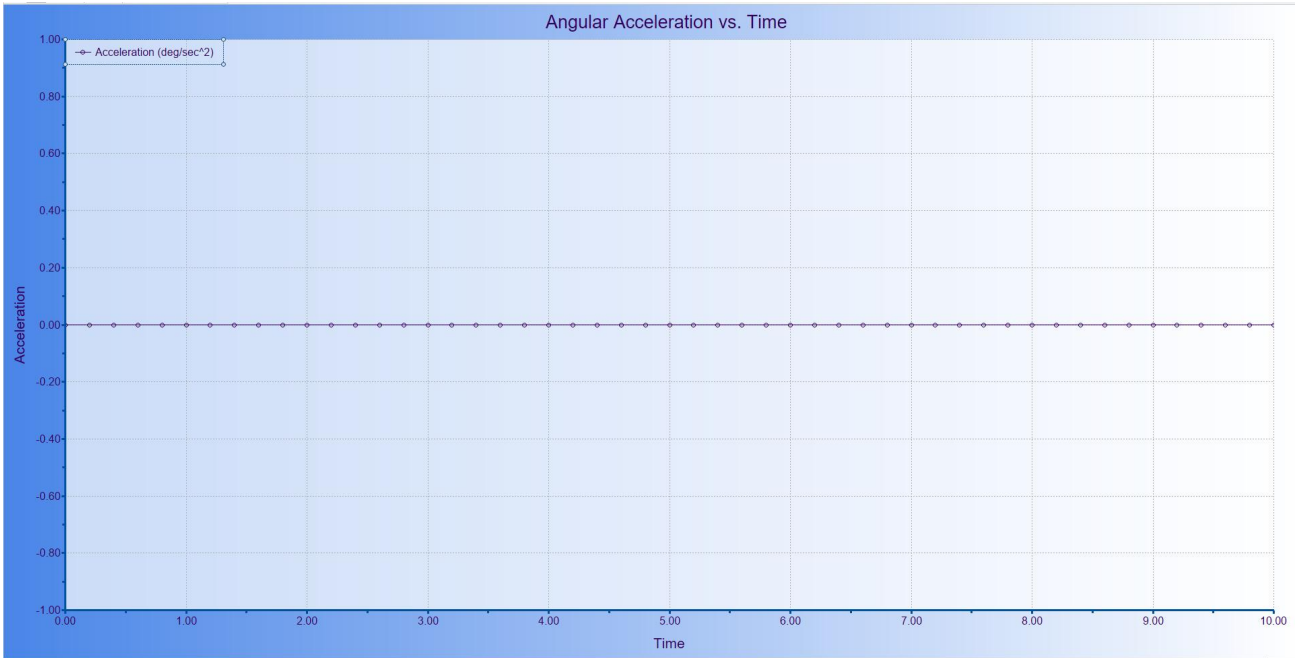


Figure 3: Angular Acceleration vs. Time Graph (Generated from Creo Mechanism)

7. Gear Ratio Calculation

For a standard planetary gear system with the ring gear fixed, the gear ratio is calculated as:

$$\text{Gear Ratio} = 1 + (N_{\text{ring}} / N_{\text{sun}})$$

Where:

- N_{ring} = Number of teeth on the Ring Gear
- N_{sun} = Number of teeth on the Sun Gear

Substituting the values from the modeled components:

$$\text{Gear Ratio} = 1 + (N_{\text{ring}} / N_{\text{sun}}) = [\text{Enter Value}]$$

The simulation results confirm this theoretical gear ratio, as the measured output angular velocity of the carrier equals the input velocity divided by the calculated gear ratio.

8. Observations & Discussion

- The planetary gear mechanism was successfully assembled and simulated in Creo Parametric without any interference between components.
- The ring gear remained stationary throughout the simulation, confirming the fixed-ring planetary configuration.
- The three planet gears demonstrated compound motion — rotating about their own axes (spin) while simultaneously revolving around the central sun gear axis (revolution).
- The planet carrier (runner_gear) exhibited smooth, uniform angular motion, consistent with a constant input speed applied to the sun gear.
- Angular velocity and position results align with the theoretical gear ratio predictions.
- The acceleration traces confirm steady-state motion with minimal transient effects post initialization.

9. Conclusion

This task successfully demonstrated the design, assembly, and kinematic simulation of a Planetary Gear Mechanism using Creo Parametric. All four components — main_gear.prt (sun/planet), ring_gear.prt, runner_gear.prt (carrier), and gear_connector.prt — were modeled and assembled with precise constraints.

The motion simulation provided clear visual and graphical insights into the angular position, velocity, and acceleration behavior of the mechanism. The results validated the theoretical gear ratio and confirmed the correct kinematic operation of the planetary gear system.

This exercise has strengthened practical understanding of mechanism design, gear kinematics, and the use of Creo Mechanism for industrial-grade motion analysis — skills directly applicable to automotive transmissions, robotic joints, and aerospace actuator systems.

10. References

- PTC Creo Parametric — Official Documentation & Help Guide
- Shigley's Mechanical Engineering Design — Budynas & Nisbett
- Theory of Machines — S.S. Rattan
- CodeAlpha Internship Task Brief — Mechanical Design & Simulation